

$$\begin{aligned} &?? \\ &?? \\ &\mathbf{X}_{\text{in}} \\ &\mathbf{X}_{\text{in}} \\ &S(\cdot) \\ &\mathbf{y} = \\ &S(\mathbf{X}_{\text{in}}) \\ &\text{norm}(\cdot) \\ &\tilde{\mathbf{y}} = \\ &\text{norm}(\mathbf{y}) \\ &U(\cdot) \\ &\tilde{\mathbf{y}} \\ &\mathbf{x}_{\text{proto}}^* = \arg \max_{\mathbf{x}_{\text{proto}}} U(\text{norm}(S(\mathbf{X}_{\text{in}}))) , \end{aligned}$$

$$\begin{aligned} (1) \quad &\mathbf{x}_{\text{proto}} \\ &\mathbf{x}_{\text{proto}} = \\ &[a, \mathbf{x}_p^\top]^\top \\ &q \in \\ &\mathcal{P} \\ &\mathbf{x}_p \in \\ &\mathcal{X}_p \\ &\mathcal{X}_a \\ &q \\ &?? \\ &overview[t][width = \\ &0.95]figure/chap5/model1.pdf \\ &overview_{\text{in}} \\ &\mathbf{x}_{\text{common}} \\ &\mathbf{x}_{\text{traffic}} \\ &\mathbf{x}_{\text{proto}} \\ &\mathbf{m} \\ &\mathbf{x}_{\text{common}} \\ &\mathbf{x}_{\text{common}} = \\ &[n, T]^\top \\ &n \in \\ &\mathbb{N}^+ \\ &T \in \\ &\mathbb{N}^+ \\ &\mathbf{x}_{\text{traffic}} \\ &\mathbf{x}_{\text{traffic}} = \\ &[\hat{\lambda}, \text{BI}_{\text{net}}, \text{BI}_{\text{node}}, \Psi_{\text{net}}, \Psi_{\text{node}}]^\top \\ &\mathbf{A} \in \\ &\{0, 1\}^{n \times T} \\ &A_{i,t} \\ &i \\ &t \\ &X(t) = \\ &\sum_{i=1}^n A_{i,t} \\ &t \\ &\mathcal{T}_i = \\ &\{t : \\ &A_{i,t} = \\ &1\} \\ &\{\Delta t_i^{(k)}\} \\ &\Delta t_i^{(k)} \\ &i \\ &k \\ &k + \\ &1 \\ &\mathbf{x}_{\text{traffic}} \end{aligned}$$

$$\begin{aligned} \hat{\lambda} \quad & \hat{\lambda} = \frac{1}{nT} \sum_{t=1}^T \sum_{i=1}^n A_{i,t} \\ \text{BI}_{\text{net}} \quad & \text{BI}_{\text{net}} = \frac{\text{ID}_{\text{net}}-1}{\text{ID}_{\text{net}}+1}, \text{ID}_{\text{net}} = \frac{\text{Var}(X(t))}{E[X(t)]} \\ \text{BI}_{\text{node}} \quad & \text{BI}_{\text{node}} = \frac{\overline{\text{CV}}-1}{\overline{\text{CV}}+1}, \overline{\text{CV}} = \frac{1}{n} \sum_{i=1}^n \frac{\sqrt{\text{Var}(\Delta t_i)}}{E[\Delta t_i]} \\ \Psi_{\text{net}} \quad & \Psi_{\text{net}} = \max_{1 \leq \tau \leq \tau_{\text{max}}} \frac{R(\tau)}{R(0)}, R(\tau) = \\ & \frac{1}{T} \sum_{t=1}^{T-\tau} \tilde{X}(t) \tilde{X}(t+\tau) \\ \Psi_{\text{node}} \quad & \Psi_{\text{node}} = \frac{1}{n} \sum_{i=1}^n \max \left(0, 1 - \frac{\text{Var}(\Delta t_i)}{(E[\Delta t_i])^2} \right) \end{aligned}$$

$$\begin{aligned} &\mathbf{x}_{\text{proto}} \\ &\mathcal{P} = \\ &\{a_1, a_2, a_3\} \\ &q \in \\ &\mathcal{P} \\ &\mathcal{X}_p \\ &\mathbf{x}_{\text{proto}} \in \\ &R^{d_{\text{max}}} \\ &d_{\text{max}} = \\ &1 + \\ &\max_{a \in \mathcal{P}} |\mathcal{X}_p| \\ &q \in \\ &\mathcal{P} \\ &q \\ &M \\ &\mathbf{x}_{\text{proto}} = \\ &[a, q, M, \dots]^\top \\ &\mathbf{x}_{\text{proto}} \\ &\mathbf{m} \in \\ &\{0, 1\}^d \end{aligned}$$